Experiment No.: - 04

Date:

Aim: LPC2148UART Interfacing LPC2148 in embedded system

Objectives: Students should understand interfacing of UART with ARM.

Requirements: Educational practice board for ARM7 LPC2148, +9V Power supply, USB A to B type cable, 20 pin Flat cable, Single lead wires, PC, Eclipse IDE, Flash Magic Utility.

Theory:

The ARM7 LPC2148 Primer board is specifically designed to help students to master the required skills in the area of embedded systems. The kit is designed in such way that all the possible features of the microcontroller will be easily used by the students. The kit supports in system programming (ISP) which is done through serial port. NXP's ARM7 (LPC2148), ARM Primer Kit is proposed to smooth the progress of developing and debugging of various designs encompassing of High speed 32-bit Microcontrollers.

UART

UART (Universal Asynchronous Receiver Transmitter) are one of the basic interfaces which provide a cost effective simple and reliable communication between one controller to another controller or between a controller and PC.

Interfacing UART Fig. 1 shows how to interface the UART to microcontroller. To communicate over UART or USART, we just need three basic signals which are namely, RXD (receive), TXD (transmit), GND (common ground). So to interface UART with LPC2148, we just need the basic signals.

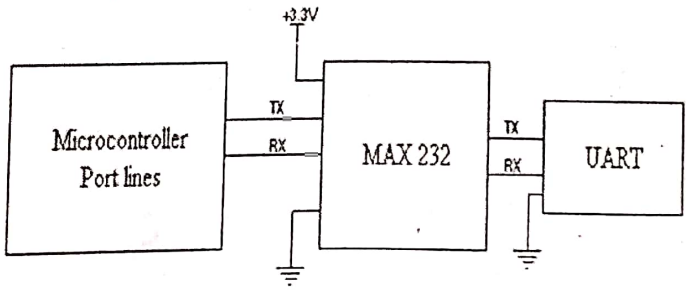


Fig. 1 Interfacing UART to Microcontroller

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Interfacing UART with LPC2148

Now want to display a text in PC from LPC2148 Primer Board by using UART module. In LPC2148 Primer Board contains two serial interfaces that are UART0 & UART1. Here we are using UART0. The Transmitter pins send the data into PC and the receiver pin receives the data from PC. The PC and microcontroller speed are denoted by using baud rate. When the baud rates of both PC and Microcontroller are same, then only the data transmit and receive correctly otherwise not.

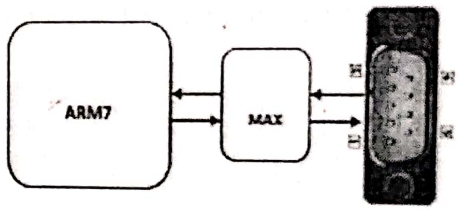
	UART DB-9 Connector	LPC2148 Processor Lines	Serial Port Section
UART0(P1) ISP PGM	TXD-0	P0.0	
	RXD-0	P0.1	
UART1 (P2)	TXD-1	P0.8	
	RXD-1	P0.9	

Fig.2 Pin Assignment with LPC2148

The Interfacing UART with LPC2148 program is very simple and straight forward, which display a text in PC from LPC2148 Primer Board through UART0. Some delay is occurring when a single data is sent to PC. C programs are written in Keil software. The baud rate of microcontroller is 9600.

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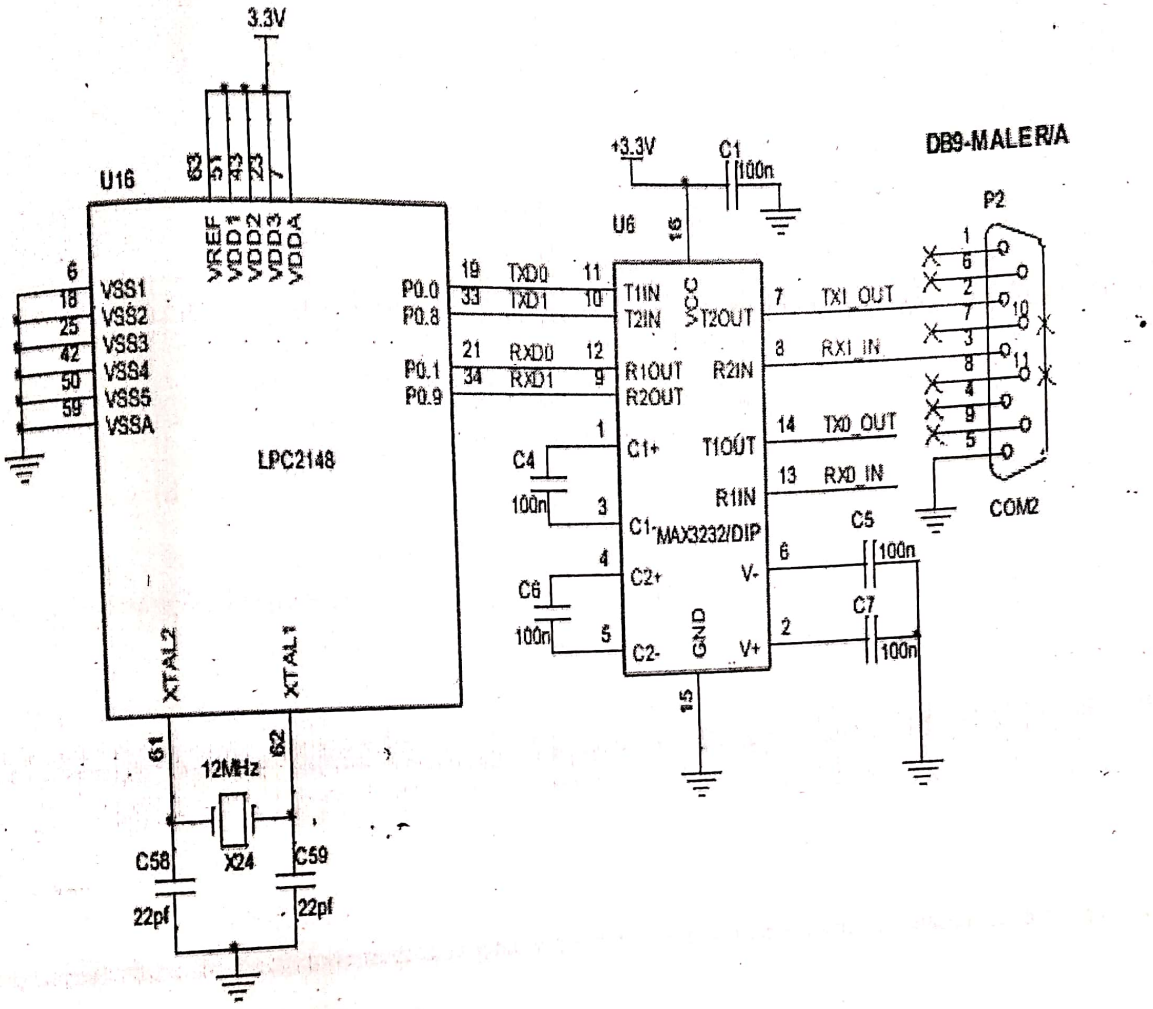


Fig.3 Interface UART0 with LPC2148

Testing the UART0 with LPC2148

Give +3.3V power supply to LPC2148 Primer Board; the serial cable is connected between the LPC2148 Primer Board and PC. Open the Hyper Terminal screen, select which port you are using and set the default settings. Now the screen should show some text messages.

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Procedure:

- **Steps to create project and program compilation:**
  - Open Eclipse.exe from –**C:\eclipse**ll folder.
  - Now browse to the ARM7 Workspace under **E Workspace** (in this case C:\E_Workspace\ARM7). Click OK to continue.
  - Click File > **Import…**, to import template.
  - Click General > Existing Projects into Workspace and click Next.
  - First Select Root Directory of your Eclipse Template for ARM7. Then select –Copy projects into workspacel check box and Click Finish.
  - Every time you import a project make sure to rename it. So **Right Click Project > Rename** or press **F2** while selecting project to rename it.
  - Go to File> New > Source File and you will see New Source File Wizard. Enter Source File name (For example **main.c**) then Click Finish.
  - Write your code and then save your Files.
  - Copy necessary .c and .h files to your local project folder. They will be added to your project in Eclipse IDE.
- **Steps to use hardware:**
  - Connect 9V DC Power supply to the educational practice board for ARM.
  - Connect the board with the USB port of the PC using the USB A to B type cable.
  - Using the RUN/PROGRAM mode selection switch, set the board in the program mode. This will be indicated by the red LED.
  - Apply Reset condition by pressing the RESET switch to ensure proper communication.
  - Using download tool (Flash Magic) download the .HEX file to the target board.
  - Connect the PL3 connector of EPB ARM7 board to PC for HyperTerminal.
  - Connect UART0 port of LPC2148 PC for all purpose kit.
  - Using the RUN/PROGRAM mode selection switch, set the board in the run mode. This will be indicated by the green LED and apply reset to execute the program.

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Code:

```
#define CR 0x0D
#include <LPC21xx.H>
void init_serial (void);
int putchar (int ch);
int getchar (void);
unsigned char test;
int main(void)
{
 char *Ptr = "*** UART0 Demo ***\n\n\rType
Characters to be echoed!!\n\n\r";
 VPBDIV = 0x02; // Divide Pclk by two
 init_serial();
 while(1)
 {
 while (*Ptr)
 {
 putchar(*Ptr++);
 }

 putchar(getchar()); // Echo terminal
 }
}
void init_serial (void)
{

 PINSEL0 = 0x00000005; // Enable RxD0 and TxD0
 U0LCR = 0x00000083; //8 bits, no Parity, 1 Stop bit
U0DLL = 0x000000C3; //9600 Baud Rate @ 30MHz VPB Clock
U0LCR = 0x00000003;
}
int putchar (int ch)
{
 if (ch == '\n')
 {
 while (!(U0LSR & 0x20));
 U0THR = CR;
 }
 while (!(U0LSR & 0x20));
 return (U0THR = ch);
}
int getchar (void)
{
 while (!(U0LSR & 0x01));
 return (U0RBR);
}
```


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Conclusion:

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Questions

1. Explain UART communication.
2. Which signals are used to interface UART with LPC2148.
3. List the Features of UART0.
4. What is the Difference between UART0 and UART1
5. Enlist the SFRs related to the UART module in LPC2148.Explain U0LCR Register in detail.

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